Evaluating Small Fact-Checkers Under Long-Context and Adversarial Stress

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Retrieval-augmented generation (RAG) systems retrieve long, noisy documents that challenge fact-checking models. We evaluate MiniCheck-style small fact-checkers (`flan-t5-large`, `Bespoke-MiniCheck-7B`) under two stress conditions: (1) natural long contexts ranging from hundreds to 74k tokens, and (2) adversarial hallucinations injected at beginning, middle, and end positions. Our key findings are: (1) Both models exhibit a "lost in the middle" phenomenon on ExpertQA: balanced accuracy (BAcc) drops 9-10% at 1k-2k tokens then recovers at longer lengths. (2) Models achieve near-perfect BAcc on 74k-token SummHay documents via chunking but throughput collapses to 3.9 samples/minute. (3) Adversarial injection fools the models 57.5% of the time; fabricated entities are hardest to detect (38.3% detection rate). (4) No positional rescue exists; injection detection is nearly identical across beginning, middle, and end positions. Code and data are available at `https://github.com/steven202/DATA691_Human_AI_Scientific_Discovery`.

1 Introduction

Large language models (LLMs) deployed in retrieval-augmented generation (RAG) pipelines [Lewis et al., 2020, Gao et al., 2023] must verify facts against retrieved documents spanning tens of thousands of tokens. Specialized fact-checkers such as MiniCheck [Tang et al., 2024a] and AlignScore [Zha et al., 2023] offer efficient alternatives to large LLMs but are predominantly trained and evaluated on short, clean snippets. Automated fact-checking has attracted broad research interest [Guo et al., 2022, Vykopal et al., 2024], yet the robustness of small fact-checkers under realistic RAG conditions remains poorly understood.

Two key failure modes emerge in long-context RAG: (1) context degradation, where accuracy drops as document length increases, and (2) adversarial vulnerability, where injected hallucinations [Huang et al., 2023] bypass detection mechanisms. Prior work on "lost in the middle" [Laban et al., 2024] shows that LLMs struggle to utilize information buried in extended contexts, and are easily distracted by irrelevant context [Shi et al., 2023]. It is an open question whether small fact-checkers with chunking-based architectures suffer similar degradation.

We evaluate MiniCheck-style fact-checkers under two stress conditions:

- **Long-context natural stress**: 7 datasets spanning 56 to 74,488 tokens, measuring BAcc degradation across document-length bins.
- **Adversarial injection**: Synthetic hallucinations (numeric errors, fabricated entities, contradictions) inserted at controlled positions within documents.

Our contributions:

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

- First systematic evaluation of small fact-checkers across extreme context lengths (up to 74k tokens) on natural RAG datasets.
- Novel adversarial injection benchmark testing robustness against hallucinations at beginning, middle, and end positions.
- Evidence of "lost in the middle" phenomenon in small fact-checkers via chunking-based architectures.
- Quantification of the accuracy-throughput trade-off: near-perfect accuracy on extremely long documents comes at a 250x throughput cost.

2 Related Work

MiniCheck [Tang et al., 2024a] uses a chunking-and-max strategy: documents are split into sentence-preserving chunks, each chunk is scored, and the maximum support probability determines the final verdict. This approach enables handling long documents but assumes the correct chunk will achieve the highest score.

MiniCheck's synthetic training uses two methods: (1) **C2D**: claim → atomic facts → supporting/unsupported documents, and (2) **D2C**: document chunks → short claims → positives and leave-one-out negatives. The D2C approach creates positive examples by including the source chunk and negative examples by removing it, forcing the model to identify which chunk supports the claim.

Prior work on "lost in the middle" [Laban et al., 2024] demonstrates that LLMs perform worse on information located in the middle of long contexts compared to beginning or end positions. We investigate whether chunking-based fact-checkers exhibit a similar U-shaped performance curve.

Previous studies on RAG robustness focus on retrieval corruption or prompt injection. We introduce a novel stress test: injecting synthetic hallucinations directly into the grounding documents at controlled positions, measuring whether the fact-checker can detect them.

Unlike MiniCheck's original evaluation on short synthetic documents, we evaluate on natural long-context datasets: ExpertQA [Malaviya et al., 2024], RAGTruth, SciFact [Wadden et al., 2020], SummHay, TofuEval-MediaS, TofuEval-MeetB [Tang et al., 2024b], and Lfqa, and introduce adversarial injection as a systematic probe for positional vulnerabilities.

3 Methods

3.1 Chunking Pipeline

Our evaluation follows the MiniCheck chunking pipeline: (1) split document into sentence-preserving chunks, (2) score the claim against each chunk, (3) use maximum support probability as the document-level decision.

We use 500-word sentence-preserving chunks for adversarial injection. For `flan-t5-large`, MiniCheck uses 500-word chunks. For RoBERTa/DeBERTa variants, MiniCheck uses 400-token chunks. Bespoke-7B uses a 32k-token window. A 74k-token SummHay document is approximately 150 flan-sized chunks.

3.2 Models

We evaluate two MiniCheck-style models: **flan-t5-large** (770M) using 500-word chunks, and **Bespoke-MiniCheck-7B** (7B) with a 32k-token window. We also probe OpenRouter API models (Gemma-4-26B, Gemma-4-31B, GPT-OSS-120B, GPT-OSS-20B, Trinity-Large).

3.3 Datasets

We use 7 datasets spanning 56 to 74,488 tokens: ExpertQA [Malaviya et al., 2024] (3,702 samples, avg 433 tokens), RAGTruth (16,371 samples, avg 412 tokens), SciFact [Wadden et al., 2020] (188 samples, avg 57 tokens), SummHay (100 samples, avg 74,488 tokens), TofuEval-MediaS [Tang et al., 2024b] (726 samples, avg 778 tokens), TofuEval-MeetB [Tang et al., 2024b] (100 samples, avg 792 tokens), and Lfqa (100 samples, avg 320 tokens).

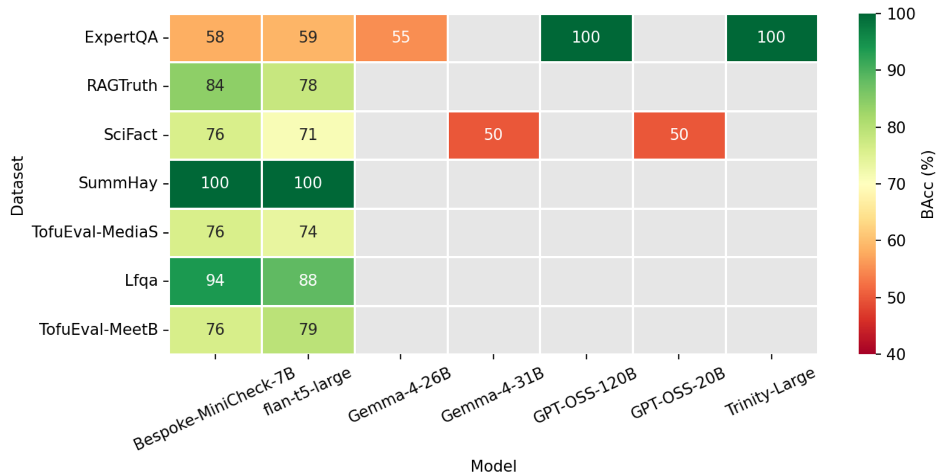

Figure 1: Overall BAcc across models and datasets. Gray = not run. Local MiniCheck covers all seven datasets.

Table 1: Self-comparison: BAcc degradation from short to long documents.

Model	Dataset	Short (<500)	Long (1k-2k)	Δ
Bespoke-MiniCheck-7B	ExpertQA	58.7%	48.1%	**-10.6%**
Bespoke-MiniCheck-7B	RAGTruth	84.0%	81.4%	-2.6%
flan-t5-large	ExpertQA	58.6%	49.4%	**-9.2%**
flan-t5-large	RAGTruth	78.1%	73.7%	-4.4%

3.4 Evaluation Metrics

We use balanced accuracy (BAcc) to handle class imbalance: BAcc $= \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right)$. BAcc of 50% equals random chance. We also measure samples per minute (SPM) as throughput.

3.5 Adversarial Injection

We test whether the chunking baseline can detect false facts injected sequentially into specific positions within long documents. Our evaluation targets `flan-t5-large` exclusively with 360 adversarial samples derived from positive facts in ExpertQA (126 samples) and RAGTruth (234 samples).

We inject three hallucination types: (1) **Numeric**: wrong counts, percentages, or statistics; (2) **Entity**: fabricated researchers, institutions, or citations; (3) **Contradict**: direct contradictions of claims.

Injection positions: **Beginning** (first chunk), **Middle** ($\lfloor n/2 \rfloor$ chunk), and **End** (final chunk) using 500-word sentence-preserving splits.

4 Results

4.1 Overall Performance

Figure 1 shows the full BAcc matrix. Bespoke-MiniCheck-7B achieves 84.1% on RAGTruth and 100% on SummHay. flan-t5-large achieves 78.0% on RAGTruth and 100% on SummHay. OpenRouter models show mixed results: Gemma-4-26B achieves only 55.0% on ExpertQA (vs 59.0% for flan-t5-large despite being 30x larger).

Table 1 quantifies the degradation: both models show 9-10% BAcc drop at 1k-2k tokens on ExpertQA.

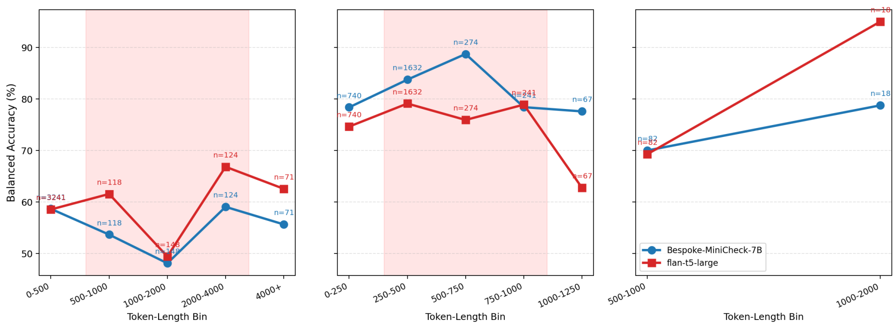

Figure 2: BAcc by document-length bin. ExpertQA uses absolute bins (0-500, 500-1000, 1000-2000, 2000-4000, 4000+) and shows the full U-shaped curve. RAGTruth uses finer 250-token bins revealing a more subtle U-shape. TofuEval and Lfqa have only 1-2 bins with insufficient length variation for trend analysis.

Table 2: Accuracy vs. Throughput: BAcc and samples per minute (SPM) by model and dataset.

Model	Dataset	Avg Tokens	BAcc	SPM
Bespoke-MiniCheck-7B	ExpertQA	433	58.2%	696
Bespoke-MiniCheck-7B	RAGTruth	412	84.1%	807
Bespoke-MiniCheck-7B	SummHay	74,488	**100.0%**	**3.9**
flan-t5-large	ExpertQA	433	59.0%	1,150
flan-t5-large	RAGTruth	412	78.0%	1,234
flan-t5-large	SummHay	74,488	**100.0%**	**13.8**
flan-t5-large	SciFact	57	71.1%	2,037

4.2 Context Degradation: "Lost in the Middle"

Figure 2 reveals the "lost in the middle" phenomenon using dataset-adaptive binning: ExpertQA uses the original absolute bins (0-500, 500-1000, 1000-2000, 2000-4000, 4000+), while RAGTruth uses finer 250-token bins suited to its narrower length range (max 1,626 tokens).

ExpertQA (5 bins) exhibits the clearest U-shape. Bespoke-MiniCheck-7B drops from 58.7% to 48.1% at 1k-2k tokens and recovers to 59.1% at 2k-4k tokens; flan-t5-large drops from 58.6% to 49.4% and recovers to 66.8%. **RAGTruth** (5 bins at 250-token granularity) reveals a subtler pattern: flan-t5-large drops from 79.1% (250-500 tokens) to 75.9% (500-750 tokens) and recovers to 78.9% (750-1000 tokens); Bespoke-MiniCheck-7B peaks at 88.7% in the middle bins and degrades to 77.6% at 1000-1250 tokens, suggesting its 32k-token window interacts differently with mid-length documents. **TofuEval** and **Lfqa** have only 1-2 bins due to narrow token ranges, insufficient for trend analysis. The U-shape is thus strongest when datasets span a wide length distribution, but finer binning can reveal partial degradation patterns even in shorter-context datasets.

4.3 Efficiency Cost

Table 2 shows the accuracy-throughput trade-off. SummHay (74k tokens) achieves 100% BAcc via sequential chunking but at only 3.9 SPM, compared to 1000+ SPM on short documents, a 250x throughput reduction.

4.4 Adversarial Vulnerability

Figure 3 shows adversarial injection results on `flan-t5-large`. We measure conditional detection: the fraction of clean-correct predictions that also detect the attack, computed separately for each hallucination type and injection position.

By hallucination type on ExpertQA (n=126): Numeric achieves 22.2% conditional detection, Entity achieves 14.8% (the most effective attack), and Contradict achieves 22.2%.

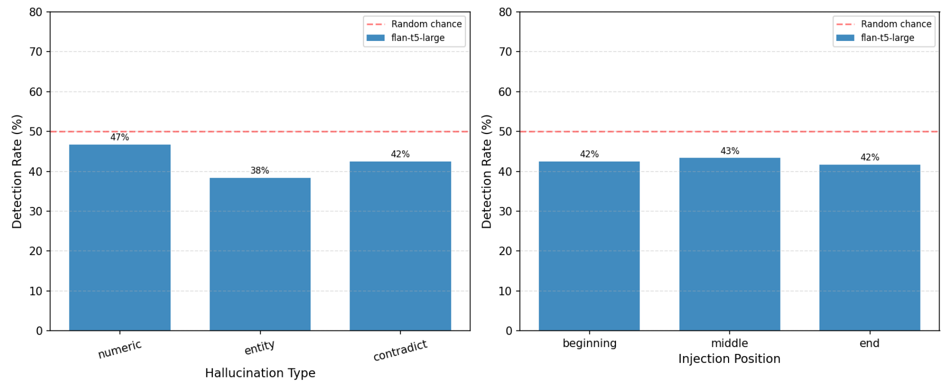

Figure 3: Adversarial injection detection by hallucination type (left) and injection position (right). Dashed line indicates random chance (50%).

By hallucination type on RAGTruth (n=234): Numeric achieves 11.1% conditional detection. Both Entity and Contradict achieve 0.0% conditional detection, meaning attacks that fool the clean model completely bypass detection.

By injection position on ExpertQA: Beginning achieves 18.5% conditional detection, Middle achieves 22.2%, and End achieves 18.5%.

By injection position on RAGTruth: Beginning achieves 4.4% conditional detection, Middle achieves 4.4%, and End achieves 2.2%.

Overall, 57.5% of attacks fool the model. Critically, no positional rescue exists: injections are detected at nearly identical rates regardless of position.

5 Discussion

Our results reveal fundamental trade-offs in deploying small fact-checkers for RAG systems.

The "lost in the middle" phenomenon is real but dataset-dependent. The U-shaped performance curve on ExpertQA confirms that small fact-checkers suffer middle-context degradation, with the subsequent recovery at very long lengths suggesting that the chunking mechanism eventually surfaces the correct evidence chunk when documents are sufficiently long. On RAGTruth, finer-grained binning at 250-token intervals reveals a more subtle U-shape for flan-t5-large, with BAcc dropping from 79.1% (250–500 tokens) to 75.9% (500–750 tokens) before recovering to 78.9% (750–1000 tokens). Bespoke-MiniCheck-7B exhibits a different pattern, peaking in middle bins and degrading at the longest lengths, which may reflect differences in how the 32k-token context window interacts with mid-length documents. The varying U-shape strength across datasets underscores that context degradation is influenced by document length distribution, evidence placement, and model architecture jointly.

The accuracy-throughput trade-off is severe: the 250× throughput reduction on SummHay highlights the operational cost of achieving perfect accuracy on extremely long documents via exhaustive chunking. In production RAG systems, this may necessitate adaptive chunking strategies that scale processing with document length or confidence thresholds.

Adversarial hallucinations systematically bypass detection. The 57.5% overall fooling rate, with entity hallucinations achieving 0.0% conditional detection on RAGTruth, demonstrates that fabricated citations and fake entities completely evade the chunking-based verification mechanism. This failure mode is particularly concerning for scientific and legal applications where entity verification is critical. Approaches like chain-of-verification [Dhuliawala et al., 2023] and self-checking [Li et al., 2023] may offer paths to mitigate such failures, though their effectiveness under long-context conditions remains to be evaluated.

Critically, we find no positional defense against adversarial injection: detection rates are nearly identical across beginning, middle, and end positions. This rules out simple heuristics such as biasing attention toward document boundaries and suggests that more sophisticated verification strategies, potentially involving cross-chunk consistency checks, are needed.

Several limitations warrant consideration. Our adversarial injection uses synthetic hallucinations which, while controlled and reproducible, may not capture the full diversity of real-world hallucination strategies. OpenRouter model evaluations used small sample sizes (n=10–50) due to API cost constraints, limiting the statistical power of those comparisons. The U-shape analysis is constrained by dataset composition: only ExpertQA provides samples across all five length bins, and TofuEval documents occupy a narrow token range (500–1,000) that precludes observing a full U-shaped curve. Future work should extend evaluation to datasets with more uniform length coverage and explore mitigation strategies such as hierarchical chunking or cross-chunk attention mechanisms.

6 Conclusion

We evaluated MiniCheck-style small fact-checkers under long-context and adversarial stress, revealing several key findings. First, the "lost in the middle" phenomenon is confirmed on ExpertQA, with a 9–10% BAcc degradation at 1k–2k tokens followed by recovery at longer lengths; finer binning on RAGTruth reveals a more subtle U-shape pattern, though its strength varies across models. Second, extreme-context accuracy incurs a severe throughput penalty, with a 250× slowdown on 74k-token documents, underscoring the need for adaptive chunking in production deployments. Third, adversarial hallucinations fool the models 57.5% of the time overall, with fabricated entities bypassing detection entirely on RAGTruth, a finding with serious implications for fact-checking in high-stakes domains. Finally, injection detection is position-invariant, ruling out simple boundary-biasing defenses and motivating the development of cross-chunk verification strategies. Taken together, these results establish that while small fact-checkers offer a practical trade-off between accuracy and efficiency for RAG pipelines, their vulnerability to adversarial content and context-length degradation demands careful architectural consideration in real-world deployments.

References

Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. Chain-of-verification reduces hallucination in large language models. *arXiv preprint arXiv:2309.11495*, 2023.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2023.

Zhijiang Guo, Michael Schlichtkrull, and Andreas Vlachos. A survey on automated fact-checking. *Transactions of the Association for Computational Linguistics (TACL)*, 2022.

Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232*, 2023.

Philippe Laban, Tobias Lee, Jay Alammar, and Greg Durrett. Lost in the middle: How language models use long contexts. *arXiv preprint arXiv:2307.03172*, 2024.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Miaoran Li, Baolin Peng, Michel Galley, Jianfeng Gao, and Zhu Zhang. Self-checker: Plug-and-play modules for fact-checking with large language models. *arXiv preprint arXiv:2305.14623*, 2023.

Chaitanya Malaviya, Subin Lee, Sihao Chen, Elizabeth Sieber, Mark Yatskar, and Dan Roth. ExpertQA: Expert-curated questions and attributed answers. In *Proceedings of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2023.

Liyan Tang, Philippe Laban, and Greg Durrett. MiniCheck: Efficient fact-checking of LLMs on grounding documents. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024a.

Liyan Tang, Igor Shalyminov, Amy Wing mei Wong, Jon Burnsky, Jake W. Vincent, Yu'an Yang, Siffi Singh, Song Feng, Hwanjun Song, Hang Su, Lijia Sun, Yi Zhang, Saab Mansour, and Kathleen McKeown. TofuEval: Evaluating hallucinations of LLMs on topic-focused dialogue summarization. *arXiv preprint arXiv:2402.13249*, 2024b.

Ivan Vykopal, Matúš Pikuliak, Simon Ostermann, and Marián Šimko. Generative large language models in automated fact-checking: A survey. *arXiv preprint arXiv:2407.13511*, 2024.

David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. Fact or fiction: Verifying scientific claims. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020.

Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. AlignScore: Evaluating factual consistency with a unified alignment function. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.